

People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research



UNIVERSITY OF ABDELHAMID MEHRI – CONSTANTINE 2

Faculty of New Technologies of Information and Communication (NTIC)

Department of Fundamental Computing and its Applications (IFA)

BACHELOR'S DISSERTATION

to obtain the diploma of Bachelor degree in Computer Science

Option: Computer Sciences (SCI)

Laboratory Application for Patient Services and AI-Driven Scan Classification

Realized by:

Benaissa Roumeissa

Hamdane Salsabil

Under supervision of:

Dr. Bounabe Rayan

June 2025

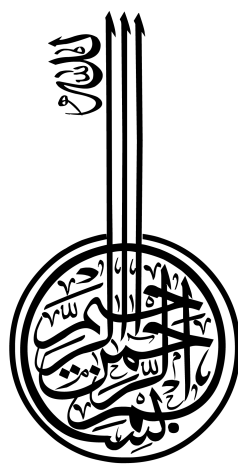


Table of Contents



Table of Contents	ii
General Introduction	1
1 State of the Art	3
1.1 Project Context and Area	3
1.1.1 Laboratory Information Management System	3
1.1.2 Toward Deep Learning for Disease Classification using Medical Imaging	4
1.2 Related Works	6
1.2.1 Comparative Table of Existing Solutions	6
1.2.2 Discussion and Analysis	7
1.2.3 General Discussion	9
2 Contribution	10
2.1 Part One: System Design	10
2.1.1 Preliminary study and requirements specification	10
2.1.2 Analysis and Design	13
2.2 Part two: Proposed Solution	16
2.2.1 General Architecture	16
2.2.2 Data Collection	17
2.2.3 Data preprocessing	19
2.2.4 Classification	21
2.2.5 model evaluation	24
2.2.6 Conclusion	26
3 Implementation & Experiments	28

3.1	Part One: Web Application	28
3.1.1	Hardware and software for web implementation	28
3.1.2	Web Application Presentation	29
3.2	Part Two: Model Experimentation and Discussion	32
3.2.1	Experimentation	32
3.2.2	Discussion	34
	General Conclusion	37
	Bibliography	38

General Introduction



Project Background

According to the World Health Organization, 1.8 million deaths are recorded from lung cancer worldwide, making it the most lethal and dangerous type of cancer. This high rate is due to several reasons and factors, the most important of which is late diagnosis of symptoms, which reduces early treatment. Therefore, we proposed this study to detect lung cancer early by expanding the scope of examinations and developing diagnostic and treatment methods through the creation of a deep learning model integrated into the website.

Problem

With the ever-increasing number of lung cancer deaths, health systems are under constant pressure to detect the disease early. Speed of diagnosis is one of the most prominent challenges facing them, as delays in diagnosis lead to unacceptable results. To reduce this problem and speed up medical diagnosis while maintaining accuracy, we have relied on artificial intelligence and deep learning.

Proposed Solutions

This study aims to contribute to improving the early detection of lung cancer by leveraging artificial intelligence and deep learning techniques to analyze medical images. In this work , we propose a CNN deeplearning modele with a High preprocessing techniques integrated into medical website .

Document Plan

In the first chapter, we will present a state of the art that includes some definitions related to laboratories and deep learning, followed by a review of related work and a discussion of their contributions. The second chapter is divided into two parts: Part One presents the website and their general diagramms; Part Two focuses on the proposed solution of this study, including the model and its architecture. chapter 3 presents all the results obtained after conducting the experiments, along with a comparison and analysis of these results .

State of the Art

Introduction

Lung cancer is one of the most dangerous diseases in the world, and early diagnosis is essential [1]. Deep learning is used to detect and classify the disease through a model trained on a dataset consisting of X-ray images. This model will be integrated into a website that allows users to upload images and receive results.

This chapter provides a comprehensive overview of the project context and area, including the laboratory information management system (LIMS) and the use of deep learning for disease classification using medical imaging. It also includes related works, presenting a comparative table of existing solutions and a discussion and analysis. Finally, it concludes with a comprehensive summary of the chapter.

1.1 Project Context and Area

This project addresses the integration of the Laboratory Information Management System (LIMS) with deep learning techniques to improve disease diagnosis through medical image analysis this integration contributes to faster processing

1.1.1 Laboratory Information Management System

Laboratory Information Management System (LIMS) is a program specifically designed for laboratories to manage and organize various operations. It helps in managing and tracking analyses while recording and saving results, ensuring high efficiency in operations. It is designed considering the need of analytical laboratories to carry out the research in fast, efficient and transparent manner with better accessibility of the instruments [2]. The program handles large quantities of samples and data daily and performs the following tasks:

- Sample management, where it tracks the status of the sample from the time of its receipt.
- Management of analyses and results including recording and entering analyses while specifying the tests.
- Storing results after examining the samples and extracting the results, they are stored according to laboratory standards.
- Patient and user management by defining roles and permissions.

1.1.2 Toward Deep Learning for Disease Classification using Medical Imaging

Deep learning is a branch of artificial intelligence that relies on deep neural networks (DNN) to simulate the way the human brain functions. It can be used to process data by analyzing medical images to diagnose and classify diseases such as pneumonia and lung cancer [3]. Lung cancer is the second most prevalent cancer in the world, and it is the most common cancer among men and the second most common among women. Accurate diagnosis requires staging of the cancer, which helps in making decisions about appropriate treatment options such as surgery, chemotherapy, radiation, and targeted therapy. Lung cancer is divided into two main types: non-small cell lung cancer (NSCLC) and small cell lung cancer (SCLC)[1].

Deep learning techniques can classify diseases (whether benign or malignant) based on specific patterns in images. The deep learning process involves several stages, such as data collection, preparation, model analysis, and system evaluation, by determining whether the images indicate a specific disease, such as pneumonia or lung cancer[4] .

Stages of Deep Learning in Disease Classification using Medical Imaging :

- ▶ **Data Collection:** Gather relevant data (e.g., chest X-ray images for disease detection).
- ▶ **Data Preprocessing:** Cleaning: Remove noise or errors from the dataset. Normalization: Scale pixel values to a range between 0 and 1.
- ▶ **Augmentation:** Apply transformations like rotation, flipping, or cropping to increase dataset diversity and improve model performance.
- ▶ **Model Design:** Choose Architecture: Use Convolutional Neural Networks (CNNs) for image data, or Recurrent Neural Networks (RNNs) for sequence data. Define Layers: Include convolutional layers, pooling layers, and fully connected (dense) layers.
- ▶ **Model Training:** Use optimization algorithms such as Gradient Descent. Adjust model weights using Backpropagation.
- ▶ **Validation and Testing:** Evaluate the trained model on unseen data (test dataset). Use evaluation metrics such as Accuracy, Recall, F1-score, and AUC-ROC. [4]

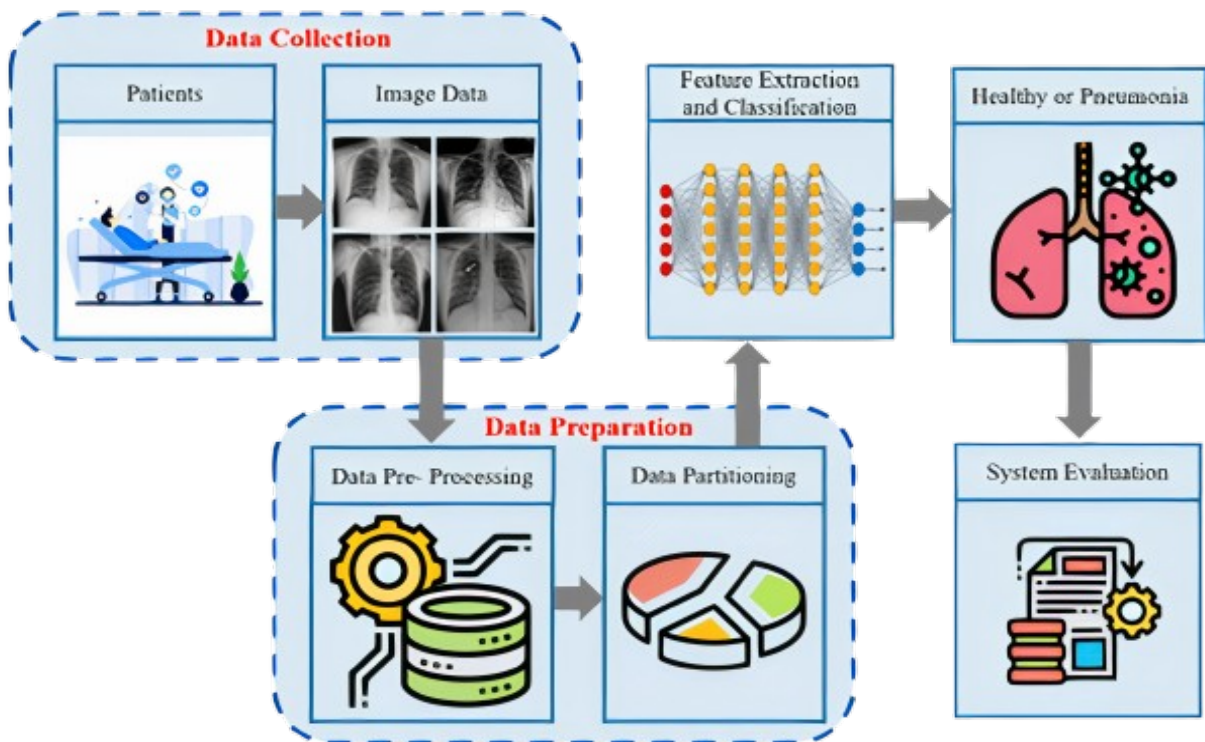


Figure 1.1: Deep Learning Workflow for Disease Detection [4]

1.2 Related Works

This section presents a some of related works on lung cancer detection using deep learning, along with a comparative analysis of the approaches and a general conclusion.

1.2.1 Comparative Table of Existing Solutions

We present here a summary table of some related works

Reference	Dataset	PREPROCESSING	TECHNIQUE	EVALUATION METRICS	RESULT
[5]	chest x-rays	Data Organization, Distribution, Annotation, Class Balancing, Data Augmentation	Mask-RCNN Avec ResNet50, ResNet101 Ensemble Model (ResNet50 + ResNet101)	Precision, Recall (Sensitivity), F1-score, Accuracy	21.81%
[6]	chest x-rays	Image segmentation, Normalization, Data Augmentation	CNNs transfer learning ensembles of multiple models	Accuracy, Precision, F1-score, Recall / Sensitivity	99.87%
[7]	X-ray (CXR) from 400×300 to 936×768	Image resizing, RGB ordering	DCNN : CJT DELT	Accuracy, Sensitivity, Specificity	DELT 97.26%, CJT 98.22%
[8]	Type : chest x-rays	Resized Image, Image normalization, Data splitting	Used two new lightweight CNN-based diagnostic models	Accuracy	98.55%, 96.83%
[9]	X-rays, WSI, CT scans, and MRI	-Image resizing	DCNN	Accuracy, Precision, Sensitivity, Specificity	0.95
[4]	X-ray images	Image resizing, Data partitioning	CNN with different models (VGG19, ...)	Accuracy, Recall	87.28%, 70.99%, 77.56%
[10]	X-ray images	Image resizing, Data Histogram Equalization Median Filtering) , normalization, Data splitting	CNN (Densenet121) with transfer learning	Accuracy, specificity , sensilivity	74.43%, 74.96%, 74.68%
[11]	X-ray images	Image resizing, Data normalization, Data splitting	model (VGG19)with CNN	Accuracy, Recall, precision,F1score	96.48%,93.75%, 97.56%,95.62%
[12]	X-ray images	Image resizing, Data blancing, standarization of pixel values of images	CNN	Accuracy	97.23%
[13]	CT-scans	Albumentations	DenseNet121	Accuracy , Precision , Recall , F1-Score	82.14%
[14]	CT scans	Image resizing, pixel normalization	VGG16, VGG19, MobileNet, InceptionV3 et DenseNet121	Accuracy (précision) ,Precision , Recall , F1-score , AUC	98.7%
[15]	CT scan	Image resizing, Normalization, Label encoding	CNN	Accuracy ,Precision , Recall F1-score	91.5%

Table 1.1: Table of Techniques and Results of CNN Models

1.2.2 Discussion and Analysis

In these papers, the authors address the issue of lung cancer diagnosis using deep learning. In this article [5], they used the Mask-RCNN Avec ResNet50, ResNet101 set Model (ResNet50 + ResNet101) technique of deep learning by inputting a dataset of chest x-rays images with Data Organization, distribution, annotation, class balancing, data Augmentation as technique of preprocessing, their work was evaluated by accuracy, precision, F1-score who reached the conclusion 21.81%. Although the results were positive, there were several limitations, such as the imbalance in the dataset, which led to inaccuracies and, in some cases, overfitting. This was due to the use of an open dataset, as the class distribution was unbalanced, and there was also insufficient data, particularly for COVID-19 cases. But In this article [6], a data set was used is chest x-rays treat with Image segmentation, Normalization, Data Augmentation using CNN modele, they achieve results 99.87%. Although the results were positive, there were several limitations, such as the imbalance in the dataset, which led to inaccuracies and, in some cases, overfitting. This was due to the use of an open dataset, as the class distribution was unbalanced, and there was also insufficient data, particularly for COVID-19 cases. In this paper [7], they used the Image resizing and RGB ordering technique of preprocessing by inputting a dataset of X-ray, CXR, and CT images with DCCN technique of deep learning based on CJT and DELT theorem the work was evaluated using accuracy, precision, sensitivity, and specificity producing optimal results with an accuracy of DELT 97.26 and CJT 98.22. Although the work is efficient, it operates within a limited academic scope, focusing on algorithms without providing information about them. Also, the imbalanced dataset leads to unreliable results. This article [9] represents addressing the topic by chest x-rays dataset with resized image, image normalization, data splitting preprocessing they use two new lightweight CNN-based diagnostic model, their results rose to 98.55 in accuracy. There are some limitations, including the reliance on open-source data, which may prevent the models from performing well in real-world clinical scenarios. Additionally, there is an imbalance in the distribution of different classes, and no comparison was made with more recent models for benchmarking. On the contrary, this article [9], they input dataset of X-rays, WSI, CT, and MRI images with Image resizing preprocessing using DCCN technique of deep learning. The work was evaluated using accuracy, precision, sensitivity, and specificity producing optimal results 0.95. Although high accuracy has been achieved, there is still room for improvement in the model. It may also face various challenges in generalization when using a new dataset. And in this paper [4] the authors use a dataset of X-ray images with image resizing and data partitioning techniques of preprocessing. They used CNN with different models (VGG19, ResNet50, InceptionV3) producing optimal results of accuracy (VGG19, ResNet50, InceptionV3 = 87.28, 70.99, 77.56 respectively).

Although the work is efficient, it does not provide a large and balanced dataset and also presents difficulties in interpreting the results. This article [10] they used deep learning techniques with a pre-trained model VGG19 on a large dataset from ImageNet, and then passed it through a CNN with 3 layers, using X-ray images along with data preparation techniques such as resizing, normalization, and random splitting. The work was evaluated based on recall, precision, F1 score, and accuracy. The model performed excellently, showing better results in classifying lung diseases (COVID-19, lung cancer). However, it has some limitations, including the use of a relatively limited dataset and the reliance on a pre-trained model, which may require additional adjustments. In this article [11], they used deep learning the CNN technique (DenseNet-121) along with transfer learning to detect cancer by inputting a general (JSRT) dataset containing X-ray images. They applied data preprocessing techniques such as histogram equalization, median filtering, image normalization, image resizing, and dataset splitting. Their work was evaluated using accuracy, specificity, and sensitivity metrics. Overall, the results were good due to their attempt to address the small dataset problem by training the pre-trained model on the ImageNet dataset. However, there are some limitations, including overfitting, low specificity, and class imbalance. In this article [12], they use the CNN model of deep learning by inputting a dataset x-rays images with image resizing, Data blancing, standarization of pixel values of images as technique of preprocessing, their work was evaluated by accuracy who reached the conclusion 97.23%. Although the obtained results are satisfactory and significant, the product still has some shortcomings. For example, it could have combined multiple deep learning models, and it could also have expanded the dataset to achieve even better results. For this paper [13], the CT images has been prepared by albumentation technique of preprocessing with DenseNet121 model of deep learning based where the work was evaluated using accuracy, precision, recall producing optimal results with an accuracy 82.14%. This study has some limitations because the model was trained on a relatively small dataset, which may limit its generalizability. It was tested only on chest CT-scan images, without validation on other imaging types or real-world clinical settings. Additionally, the study focused solely on the DenseNet121 architecture, without comparison to other deep learning models.

In this article [14], the authors use VGG16, VGG19, MobileNet, InceptionV3 et DenseNet121 the technique of deep learning by inputting a dataset of CT images with image resizing, pixel normalization as technique of preprocessing, their work was evaluated by accuracy, precision, F1-score, AUC who reached the conclusion 98.7%. The system is limited by its reliance on CT scans, sensitivity to image quality, lack of patient ID verification, and absence of integration with hospital systems. It also cannot detect cancer stages or metastasis, and does not support real-time alerts.

In this article [15] , the authors choose CT images as dataset with image resizing, normalization and label encoding as technique of preprocessing , where k was evaluated by accuracy , precision , F1-score , F1-score who reached the conclusion 91.5% accuracy of model. The system is limited by its reliance on CT scans, sensitivity to image quality, lack of patient ID verification, and absence of integration with hospital systems. It also cannot detect cancer stages or metastasis, and does not support real-time alerts.

1.2.3 General Discussion

In general, studies have introduced deep learning techniques for lung cancer diagnosis by analyzing X-ray images. Models such as CNN, DCNN, and ResNet have shown good results in disease classification. However, this is not sufficient due to challenges related to the dataset and generalization across all cases. To overcome these limitations, it is essential to select a model based on the dataset used and integrate multiple models to enhance and improve diagnostic accuracy using powerful libraries and techniques in data preparation such as augmentation and tensorflow .

Conclusion

This chapter presented the classification of one of the deadliest diseases in the world using the latest technologies of deep learning, utilizing X-ray images. The work was integrated into a website to facilitate communication and interaction between patients and specialists. It also covered definitions of Laboratory Information Management Systems (LIMS) and deep learning, followed by a review of related work and a comparative analysis of different methods.

In the next chapter , we will present the website architecture adopted in this study, followed by the CNN model construction process with its various stages.

Contribution

Introduction

To ensure good diagnosis of lung cancer and good interaction between patients and specialists, we have built a website as an interaction interface between the two parties. The chapter begins by presenting the design and structure of the website developed. It is based on UML (Unified Modeling Language) diagrams, including the use case diagram, class diagram, and activity diagram. It illustrates the system's functionality, structure, and dynamic behavior. Following the system's design, the chapter shifts focus to the development of a deep learning model for lung cancer detection. It outlines the major steps involved in building the model starting with data collection and preprocessing moving on to the construction of a deep learning model and ending with the evaluation of the model's performance. The integration of the trained model into the website ensures that users can interact with the diagnostic system in a simple and efficient way.

2.1 Part One: System Design

This section presents the structure of the web application by describing the user interface components and the overall system architecture.

2.1.1 Preliminary study and requirements specification

The platform presented in this project is a medical website, developed with the primary goal of early detection of lung cancer based on X-ray images. This website represents an innovative digital solution that combines web technologies and artificial intelligence (AI) capabilities to assist patients and healthcare professionals in making accurate and rapid diagnoses of this serious disease.

The system relies on an organized structure that includes two main sections:

Patient Section

This platform is an interface that showcases personalized healthcare services for patients in order to provide remote healthcare and facilitate early treatment of lung cancer. The services offered on the website include:

Home Page : Provides a brief overview of the website and the available services.

Contact Us : This page allows users to get in touch with the administration for any issues or inquiries.

Login and Registration : The patient can create an account using the registration form, or log in if they already have an existing account.

Scan Image : Allows the user to upload an image (with limited size and in X-ray format), which is then processed using artificial intelligence (AI), and a response is sent back.

Book Appointment : If a patient wants to book an appointment, they enter their information (full name, email address, phone number, etc.), choose the preferred date and time, and submit the form. The request is reviewed, and a notification is sent with a response.

check Appointment : Admins can respond to the appointment request by clicking the notification and either confirming or canceling the booking.

Account Upgrade : The patient can upgrade their account to access additional services, such as selecting the appropriate doctor and more.

Suggest Doctor: This service displays a list of doctors the patient can book an appointment with or contact.

Admin Section

This platform provides a dedicated interface for the administrator to manage patient data by performing tasks that ensure fast and efficient management. These tasks include:

Login: The admin uses their email and password to log in. After verification, they are granted access to the site.

Dashboard management: they provide an overview of the site and statistics on patients in comparison to the available services.

View of patient: they manage the list of patients, with the ability to access each patient's file and add or modify their information

Check appointment: Upon receiving appointment requests from patients, the admin can approve or reject them and coordinate scheduling between different patients.

Check patient inquiries: They also manage patients' messages and inquiries in order to respond to them.

This process diagram illustrates the main interactions and operations within the medical website, showing how both patients and administrators navigate through the system to perform key actions such as login, appointment booking, image scanning .

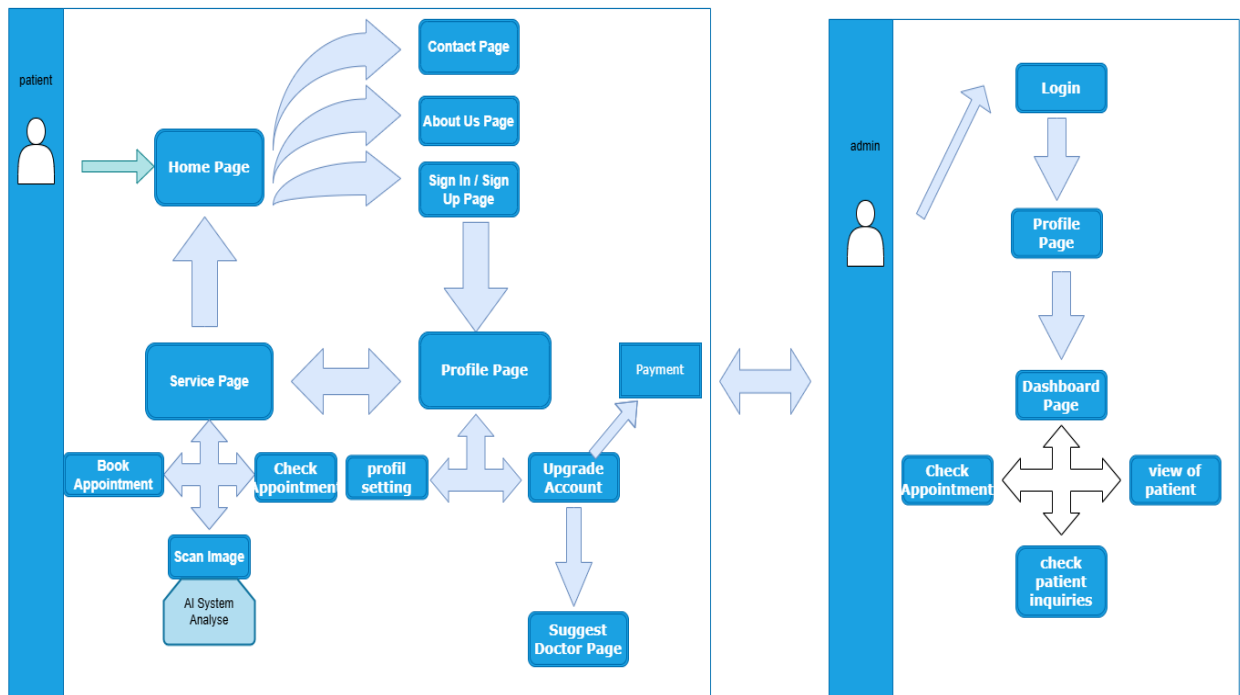


Figure 2.1: Process Diagram – Medical Website (Patient Admin)

Use Case Diagram

In this section, we present the Use Case Diagram which illustrates the main functionalities offered by the medical website and the interactions between the users and the system.

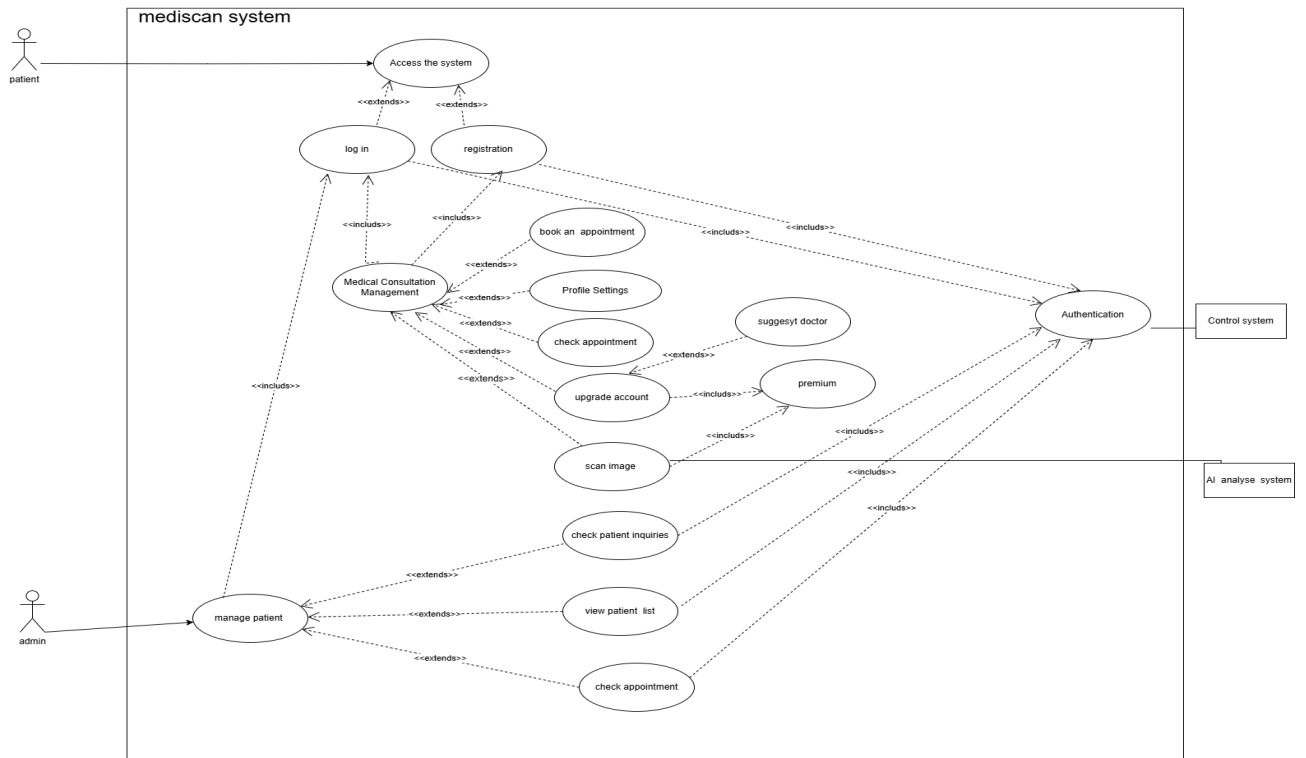


Figure 2.2: Use Case Diagram of the Medical Website System

2.1.2 Analysis and Design

UML diagrams are used to clearly define the objectives of the medical system by representing core processes and the relationships between users (such as administrator and patients) and the system, contributing to the development of an efficient platform that meets user needs effectively.

Activity Diagram

In this section, we present the Activity Diagram that illustrates the sequence of actions and processes performed by the user within the medical website.

It provides a comprehensive overview of the main user interactions with the system, starting from login and appointment booking to image scanning and doctor suggestion.

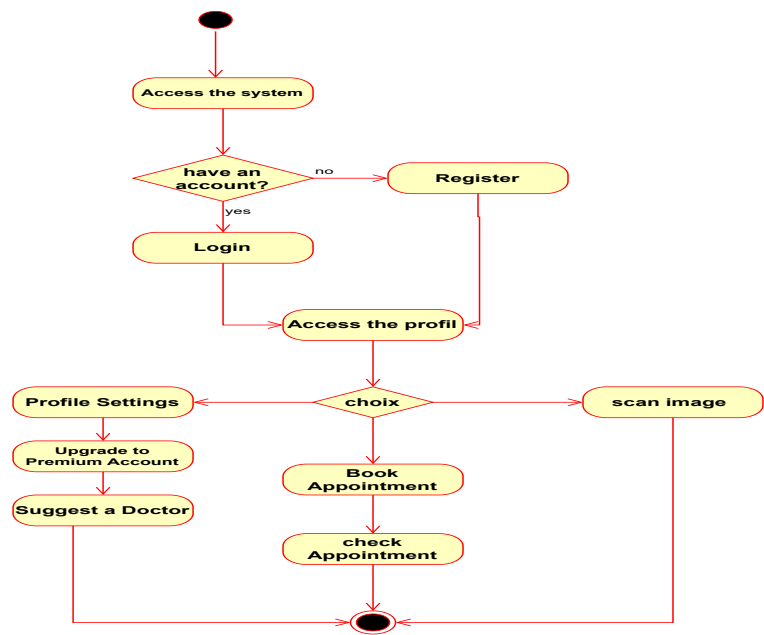


Figure 2.3: Activity Diagram – Global for the Medical Website

This activity diagram illustrates the workflow of the medical image scanning process, starting from image input to obtaining classification results using a deep learning model.

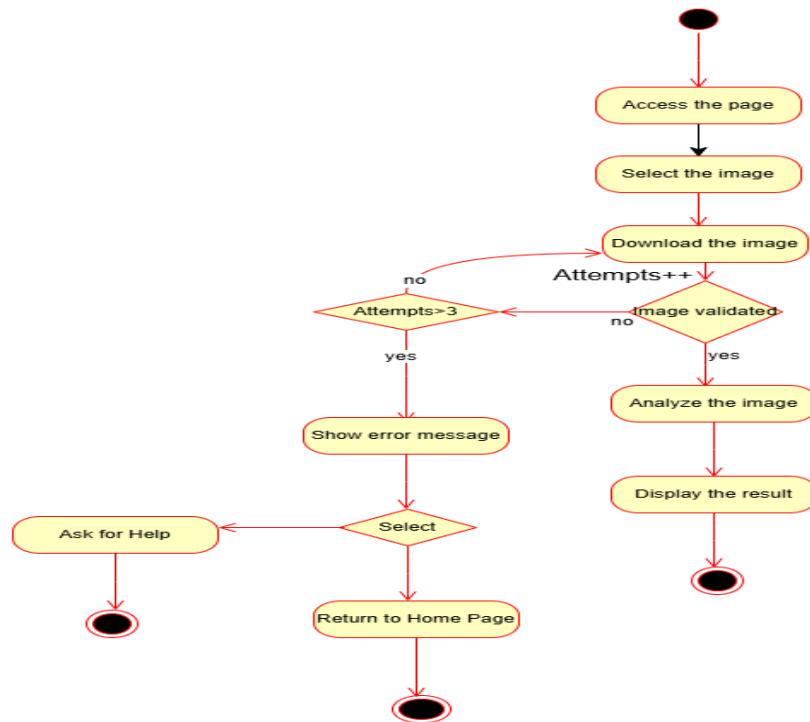


Figure 2.4: Activity Diagram – Specific to scanning image

This activity diagram illustrates the workflow of the login process, starting from writing email and password to login our system.

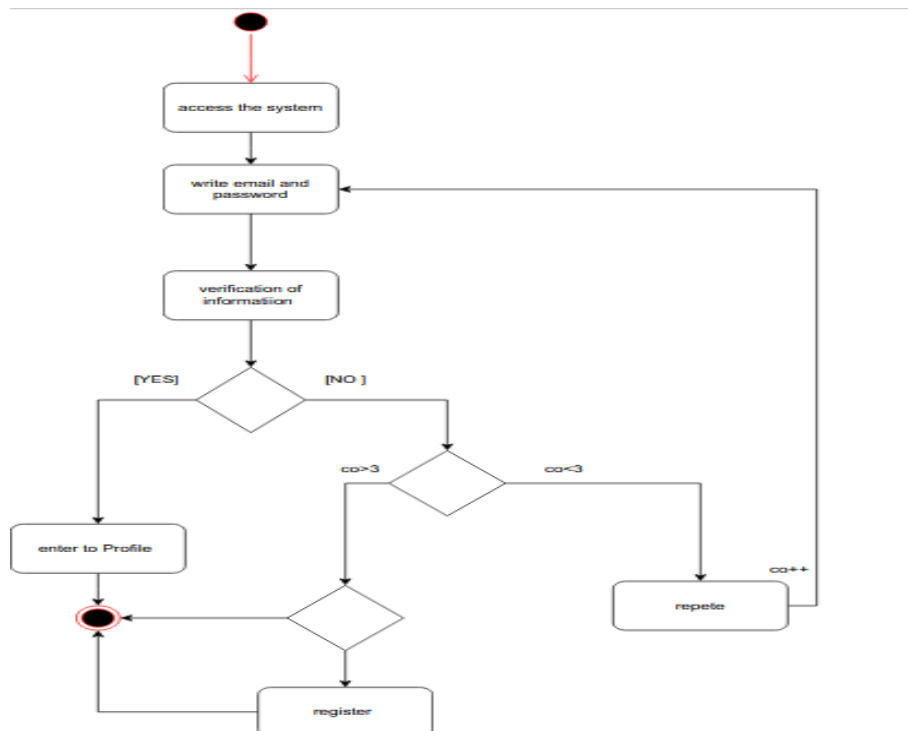


Figure 2.5: Activity Diagram – Specific to login

Class Diagram

In this section, we present the class Diagram that illustrates the defferent database of the medical website.

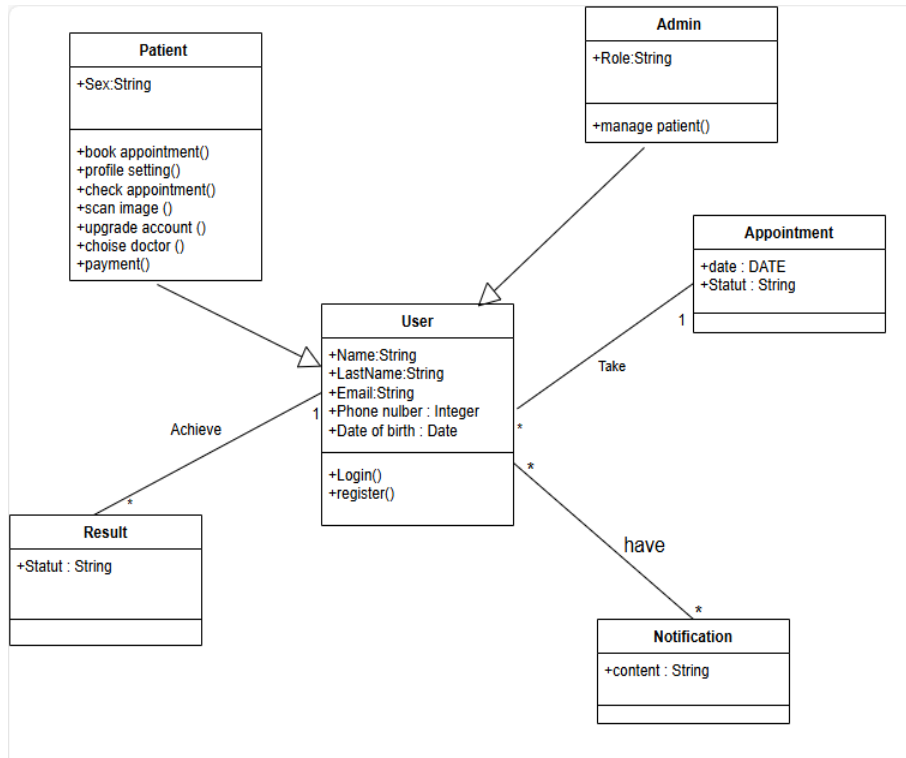


Figure 2.6: Class Diagramme

2.2 Part two: Proposed Solution

In this section, we present the deep learning process to obtain an effective model.

2.2.1 General Architecture

This global architecture illustrates all the stages of building a deep learning model .

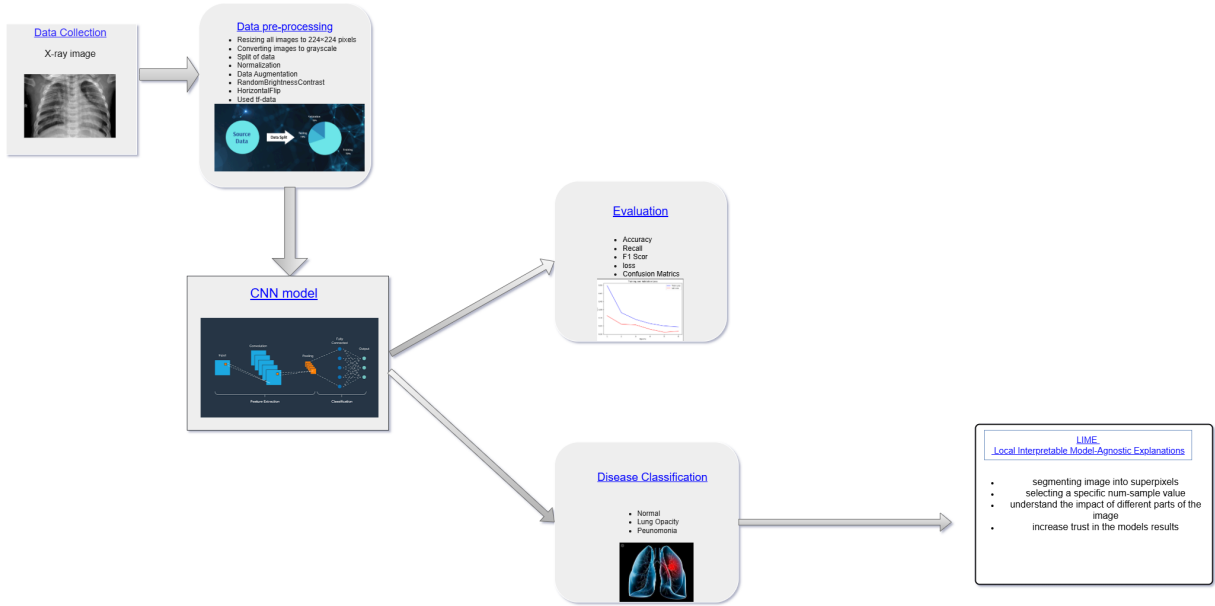


Figure 2.7: Global Architecture of process of deep learning

2.2.2 Data Collection

The data collection phase is one of the essential stages in building an intelligent system based on deep learning, especially in sensitive fields such as the medical domain. The quality and effectiveness of the data clearly determine the accuracy of disease diagnosis and the results obtained. In our work, we employed the COVID-19 Radiography Database from kaggle ¹. This dataset contains chest X-ray (CXR) images categorized into four classes: COVID-19, Normal, Viral Pneumonia, and Lung Opacity. We excluded the COVID-19 class and used only the Normal, Viral Pneumonia, and Lung Opacity classes, comprising 10,192 Normal, 1,345 Viral Pneumonia and 6,012 Lung Opacity. Although the dataset includes ground-truth lung segmentation masks, we did not use the masks in our study. To further enrich our pneumonia-related data, we added a fourth class, Pneumonia, using chest X-ray images from the Chest Xray Pneumonia dataset available at kaggle ². For our study, we extracted only the Pneumonia class, comprising 4273 images, to broaden the diversity of lung infections and enhance model training.

The compiled dataset consists of 22133 X-ray images that are accurately classified into three categories, each representing a specific condition:

Normal This class includes images of healthy individuals with no signs of lung infections or abnormalities. These images serve as a baseline for the model to differentiate between normal and pathological conditions. They are essential for training the model to accurately detect deviations from healthy lung structures.

¹Chest X-Ray Images (Pneumonia)

²COVID-19 Radiography Database

Lung Opacity This class contains images showing various forms of lung opacities . These images help the model learn to identify subtle to pronounced patterns of lung abnormalities.

Pneumonia This class includes a total of 5,929 X-ray images of confirmed pneumonia cases, collected from two different classes (Pneumonia and Pneumonia viral) to ensure diversity. The inclusion of this class enhances the model’s ability to distinguish pneumonia from other lung conditions such as general opacities thereby increasing diagnostic precision.

The images were stored in separate folders based on their category, in various formats such as JPEG, PNG, and JPG, and with different sizes across each class. Next, we will provide an image for each category.

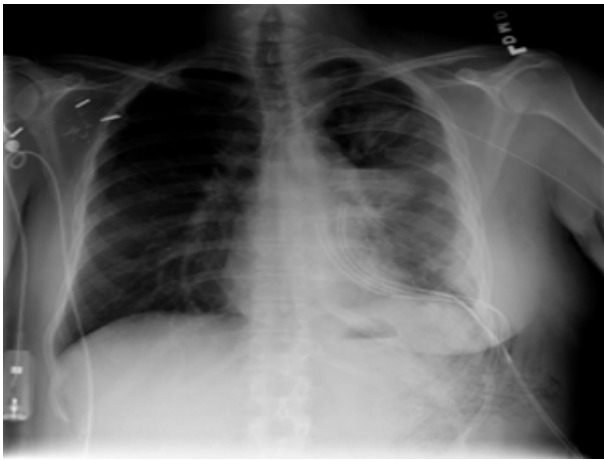


Figure 2.8: Lung Opacity



Figure 2.9: Normal

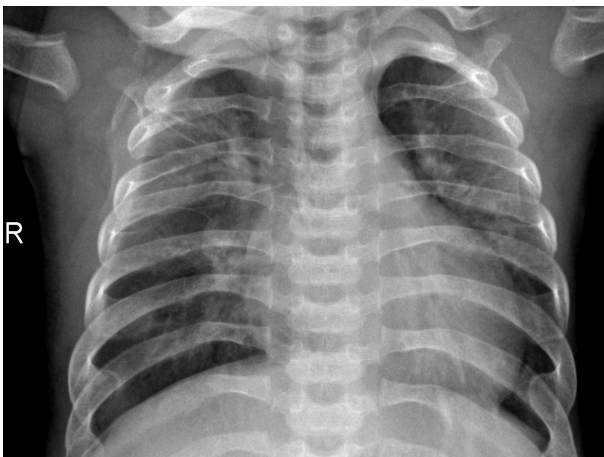


Figure 2.10: Pneumonia

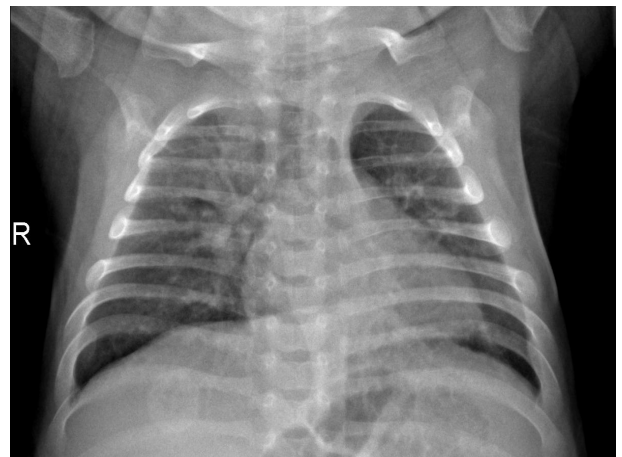


Figure 2.11: Pneumonia-Viral

2.2.3 Data preprocessing

Before training the model, preprocessing is used to classify and prepare the data in order to improve its quality and ensure better model performance. In this section, we present the preprocessing steps applied to the dataset, which consists of chest X-ray images classified into three categories: *Lung Opacity*, *Normal*, *Pneumonia(including Viral Pneumonia)*[16]. The following preprocessing steps were applied:

1. Dataset Splitting

The images were split into training (80%), validation (10%), and testing (10%) sets. We used the train-test-split technique, which includes parameters such as stratify and random-state was used to ensure a reliable evaluation of the model on unseen data, improving generalization and robustness.

2. Data Augmentation

Augmentation was applied using the Albumentations library in order to avoid overfitting and enhance the dataset. We applied some random transformations only on the training images ,such as : HorizontalFlip,RandomBrightnessContrast,Rotate,Image Resizing,Normalization .But validation and test data only use resizing and normalization techniques [13].

3. Image Resizing

The images were resized to 224×224 pixels because this size provided better results compared to other dimensions. It also helped speed up both the model training process and the testing process [13].

4. Normalization

Since pixel values are originally in the range [0, 255], they were normalized to the range [0, 1]. This improves training speed and ensures better gradient flow during backpropagation due to smaller and more uniform values [13].

5. HorizontalFlip

A technique that horizontally flips the image in order to diversify the training data and help the model generalize better [13].

6. RandomBrightnessContrast

It randomly adjusts the brightness, either by increasing or decreasing it, to help the model handle different lighting conditions and contrast [13].

7. Rotate

This technique rotates the image by a limited angle to increase the diversity of the training data [13].

8. Conversion to grayscale

We used the OpenCV library to read images in gray-scale format, meaning a single-

channel matrix representing brightness levels in black and white. Medical X-ray images are naturally black and white, and this helps the model during training. We used the method `cv2.imread` with the gray-scale flag [13].

9. Used tf-data

We used the TensorFlow library to create datasets for each stage: training, validation, and test. We applied the `tf.numpy-function` method to link operations that are not native to TensorFlow, such as OpenCV and Albumentations. We also applied batching, prefetching, and shuffling. The goal of these operations is to help load images from disk, organize them, and present them in batches, thus accelerating the loading process before sending them to the model [14].

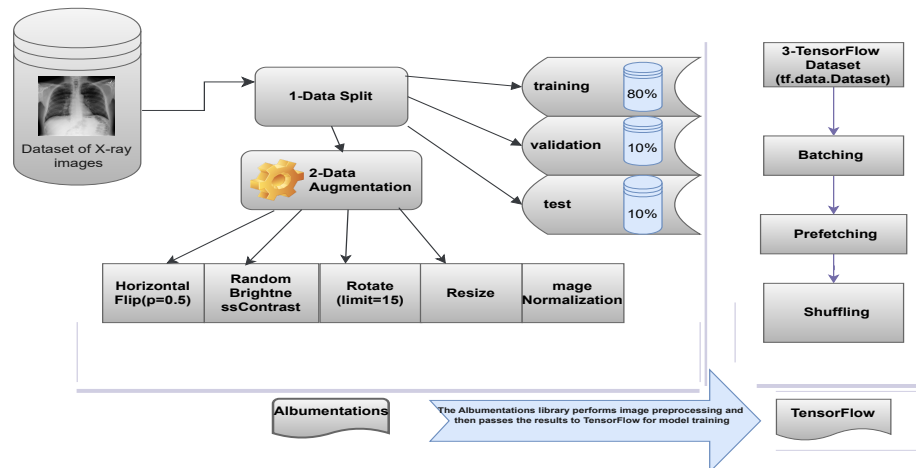


Figure 2.12: Preprocessing Techniques for Image-Based Classification

2.2.4 Classification

2.2.4.1 Model Architecture

here we present our model architecture .

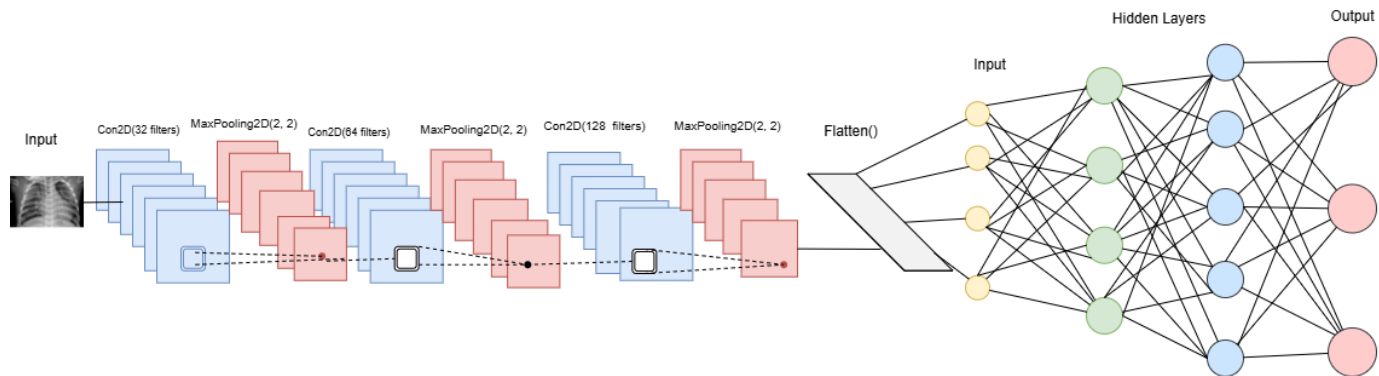


Figure 2.13: Global Architecture of process of deep learning

2.2.4.2 Model Training

To detect lung cancer using artificial intelligence techniques, a Convolutional Neural Network (CNN) deep learning model was chosen, as it is considered one of the most accurate and suitable models for diagnosing chest X-ray images. CNNs are particularly well-suited for classifying X-ray images due to both scientific and technical reasons including:

1/ High capability of automatic feature extraction : The model analyzes the image and detects patterns related to the disease (e.g., abnormal masses, changes in lung tissue) automatically without human intervention [17].

2/ Special handling of medical images: CNN treats the image as a set of pixels and applies filters to extract features [17].

3/ Prevention of overfitting: Thanks to pooling layers CNN reduces the data size progressively, which speeds up the training process and reduces the risk of overfitting [17].

The CNN model chosen in this study based on the aforementioned features includes :

Input layer : Gray-scale X-ray images with a size of 224x224 pixels are fed into the model.

Convolutional layers : The first convolutional layer applies 32 filters (small matrices for feature extraction) of size 3x3 and uses the ReLU activation function, which helps extract simple features.

The second layer has 64 filters, more complex than the first, allowing the model to detect more intricate shapes.

The third layer contains 128 filters to extract deeper and more detailed features, helping to identify patterns associated with the disease.

Pooling layers: After each convolutional layer a pooling layer is applied using MaxPooling with a size of 2x2. This reduces the image dimensions while retaining essential information which in turn reduces the number of parameters and improves processing efficiency.

Flattening layer : Converts the 3D output into a 1D vector so it can be fed into the dense layers.

Dense layer: Contains 128 units to process the combined features and discover relationships between them using an activation function to do so.

Dropout layer (rate = 0.5) : Randomly disables 50% of the neurons during training to reduce the risk of overfitting

Output layer: Contains 3 classes for classification and uses the Softmax activation function to generate probabilities indicating the likelihood of the image belonging to each class.

It has been shown that using CNNs is an accurate and effective choice for achieving better results and improved diagnosis.

2.2.4.3 Model Parameters

To improve the accuracy and effectiveness of the model and enhance the results, it is essential to fine-tune its configurations. The CNN model relies on two types of configuration: parameters and hyperparameters.

parameters : refers to a variable that the model learns from the training data. These values are automatically adjusted during the training process to minimize errors and improve performance.[3] They are mainly

- ▶ weights: the values of the filters (kernels) in convolutional layers.[3]
- ▶ Biases: additional values added to the result of a convolution to allow the network to better fit the data. These parameters are updated during backpropagation using an optimization algorithm like gradient descent.[3].

Hyperparameters : are configuration settings set by the developer before training the model. They are not learned from the data, but rather guide the learning process itself. These values control how the model is trained and can significantly affect performance and accuracy. [3]. Hyperparameters can be categorized based on their function into two main types :

- ▶ Training Hyperparameters : which controls training in the model . They is represented by [3] :

ReduceLROnPlateau : a learning rate scheduler that reduces the learning rate when a metric has stopped improving.

Batch Size : the number of training samples used to calculate one update to the model's weights during training.

Epoch : is one complete pass through the entire training dataset. Models are usually trained for multiple epochs.

Optimizer : an algorithm that modifies model weights to minimize the loss function by calculating gradients and updating weights accordingly

Loss Function : a function that quantifies how far off a model's prediction is from the true label; used during training to improve the model.

Early Stopping : a regularization technique that stops training when the model's performance on a validation set starts to degrade, preventing overfitting.

- Architecture Hyperparameters : Architecture hyperparameters are the settings that define the structure and design of a neural network which determine how the model is built. They are represented by [3] :

Layers number : The total number of layers in the model architecture, determining its depth.

Filters number : is the number of convolutional filters used in a convolutional layer, which determines how many features the model learns.

Kernel Size : the dimensions of the filter that slides over the input to extract features.

Activation Function : a non-linear function applied to each neuron's output to introduce non-linearity into the model.

Units number (in Dense layers): The number of neurons in a fully connected (Dense) layer. This controls how much information the layer can process.

Dropout rate : The fraction of neurons randomly disabled during training to prevent overfitting.

Here is a diagram that illustrates the different parameters and hyperparameters used in building and training the model.

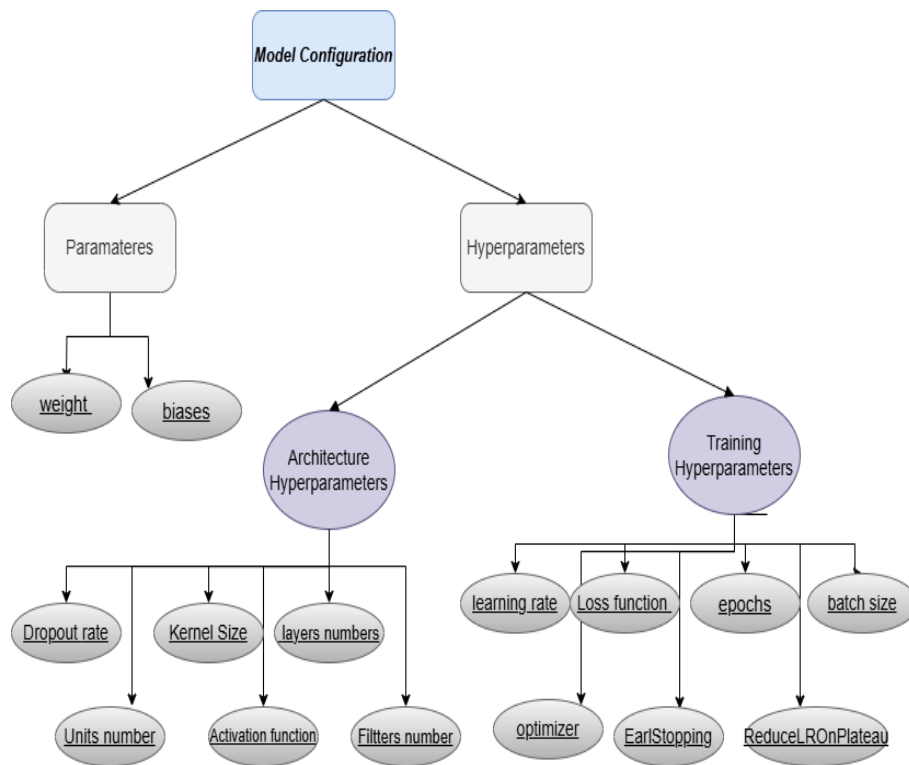


Figure 2.14: Model Configuration

2.2.5 model evaluation

After training the deep learning model, it is essential to evaluate its performance to ensure its ability to generalize to the data. There are several performance metrics, such as accuracy, recall, confusion matrix, and others. [?].

The evaluation process includes the following metrics:

1. **Accuracy:** Measures the percentage of correct predictions out of the total number of predictions. It gives a general indication of the model's performance [3].
2. **Loss:** Represents the difference between the model's predictions and the actual values. It quantifies how far the predicted values are from the ground truth. A lower loss means better performance [3].
3. **F1 Score:** The harmonic mean of precision and recall. It is a balanced metric that considers both false positives and false negatives and is particularly useful in evaluating model performance on imbalanced datasets [3].
4. **Recall:** Also known as sensitivity, recall is the ratio of correctly predicted positive observations to all actual positives (true positives + false negatives). It tells how well the model identifies actual positive cases [3].
5. **Precision:** is a classification metric used to determine the proportion of correctly predicted positive cases out of all the cases predicted as positive. It helps assess the

correctness of the predicted positive results and evaluate the model's performance [3].

6. **Confusion Matrix:** A table that visualizes the performance of the classification algorithm. It shows actual class labels versus predicted class labels. Rows represent the actual classes, and columns represent the predicted ones. It is especially useful for identifying which classes the model confuses [3].
7. **Normalized Confusion Matrix:** A version of the confusion matrix where values are normalized (i.e., expressed as percentages rather than absolute numbers), providing clearer insights into class-wise performance [3].
8. **Accuracy and Loss Curves**
 - ▶ **Accuracy Curve:** Shows the evolution of the model's accuracy at each epoch, helping to determine whether there is overfitting, underfitting, or if the model is learning properly, i.e., generalizing well to the data [3].
 - ▶ **Loss Curve:** Represents the error rate at each epoch. The lower the loss, the better the models performance [3].

Explaining CNN model decisions using LIME (Local Interpretable Model-Agnostic Explanations): the CNN model is considered a black box, meaning it makes decisions without us knowing exactly how or on what basis. For this reason, we used LIME to explain and interpret its decisions. It works by taking an image from the test dataset and segmenting it into superpixels. Since the original image has only one channel (grayscale), it is converted from RGB to grayscale. Then, we apply LIME by selecting a specific num-sample value, which defines how many perturbed (noisy) images LIME generates to understand the impact of different parts of the image. We also use top-labels to define how many top predicted classes we want to explain. After that, we use the get-image-and-mask method to highlight the most important regions (features) that had a positive impact on the models prediction. This approach helps increase trust in the models results, especially when the highlighted features are focused on the lungs, which is medically relevant [15].

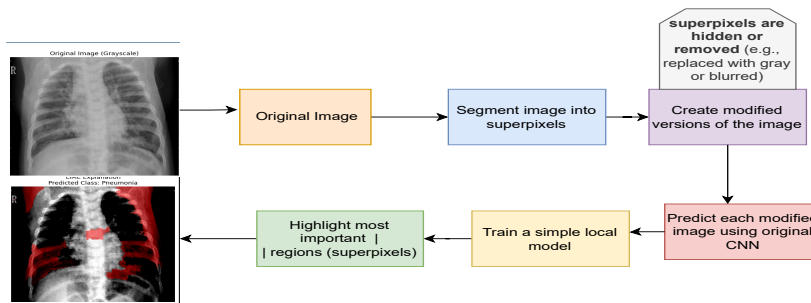


Figure 2.15: LIME Process Steps for Enhancing Interpretability

2.2.6 Conclusion

In this chapter, we presented a comprehensive overview of the medical website dedicated to the early detection of lung cancer using artificial intelligence techniques. Which focuses on the User Interface and System Architecture. This includes the general presentation of the medical website platform (Patient Section and Admin Section), as well as the UML diagrams of the system such as the Use Case Diagram, Activity Diagram, and Class

Diagram. Additionally, we discussed the deep learning approach for medical diagnosis, highlighting its main phases: general architecture, data collection, data preprocessing, deep learning model, and model evaluation.

In the next chapter , we will present the various tools used to build the website and the model and We will provide a complete explanation of the medical website and how to interact with it, along with presenting the experiments we conducted to achieve the best model.

Implementation & Experiments

Introduction

In this chapter, we present the software and hardware used in the development of this project. An overview is also provided on how the website is used and how users interact with it. Furthermore, we present the experiments conducted to reach the best model, along with a comparison between our model and previous models in this field, followed by a discussion and analysis of the results obtained. This aims to better understand the model and study its limitations in order to improve it in the future.

3.1 Part One: Web Application

To detect lung cancer using deep learning, we created this website called "mediscann", which is used by patients and others to facilitate the detection process and other related tasks.

3.1.1 Hardware and software for web implementation

We used a set of software and hardware tools to develop the Mediscann website. The details are presented in the tables below:

Table 3.1: Hardware Used for Development

Hardware	Description
Developer Device	HP (Hewlett-Packard) laptop
Processor	Intel Core i5
RAM	8 GB (Gigabyte) Random Access Memory (RAM)
Storage	256 GB SSD (Solid State Drive)
Operating System	Windows 10

Table 3.2: Software Used for Development

Software	Description
Source Code Editors	VS Code (free and open-source), PyCharm (advanced IDE for Python)
Front-end Technologies	HTML (HyperText Markup Language), CSS (Cascading Style Sheets), JavaScript
Back-end Framework	Django (Python-based framework for secure and robust applications)
Database	SQLite integrated with Django (no external database configuration needed)
Programming Language	Python 3.13.2 (interpreted, versatile, easy to read)
Additional Tools	Anaconda (distribution of Python and R with Jupyter Notebook), Kaggle (platform for datasets, notebooks, and competitions)
Libraries	os, cv2, numpy, tensorflow, sklearn.model_selection, albumentations

3.1.2 Web Application Presentation

In this section, we will provide you with a comprehensive description of our medical website and how to use it, highlighting its main features, along with showcasing some user interface screens.

This website features a simple, easy-to-use, and attractive design with calm colors that suit the medical theme. It is also a responsive site that works efficiently on various devices such as smartphones, computers, and others.

The website presents several main pages, offering an overview of its goals and available medical services, with easy access to contact the support team.

As for how to use the web application, it involves the following steps:

The user begins by logging in through a dedicated interface, where they are asked to enter their email and password.

After a successful login, the user is directed to their Profile Page.

Through this page, the user can access the Service Page, where they can book appointments, upload X-ray images, and receive diagnostic results. After booking an appointment, the user can fill out a form and submit it to receive a confirmation or rejection of the appointment, with the option to either confirm or cancel it as desired.

Additionally, users can upgrade their accounts to benefit from extra services such as "Suggest Doctor," after making a manual payment. Once the payment is verified, the additional services are activated.

The principal feature of this website is allowing users to upload a medical X-ray image through the platform, which will be analyzed using artificial intelligence (Deep Learning) techniques to determine the patient's condition and classify the type of disease.

we will showcase some screenshots of the user interfaces related to the Patient Section. In this patient management platform, the system is supervised by administrators through dedicated dashboards, where the admin logs in and is directly redirected to the main page.

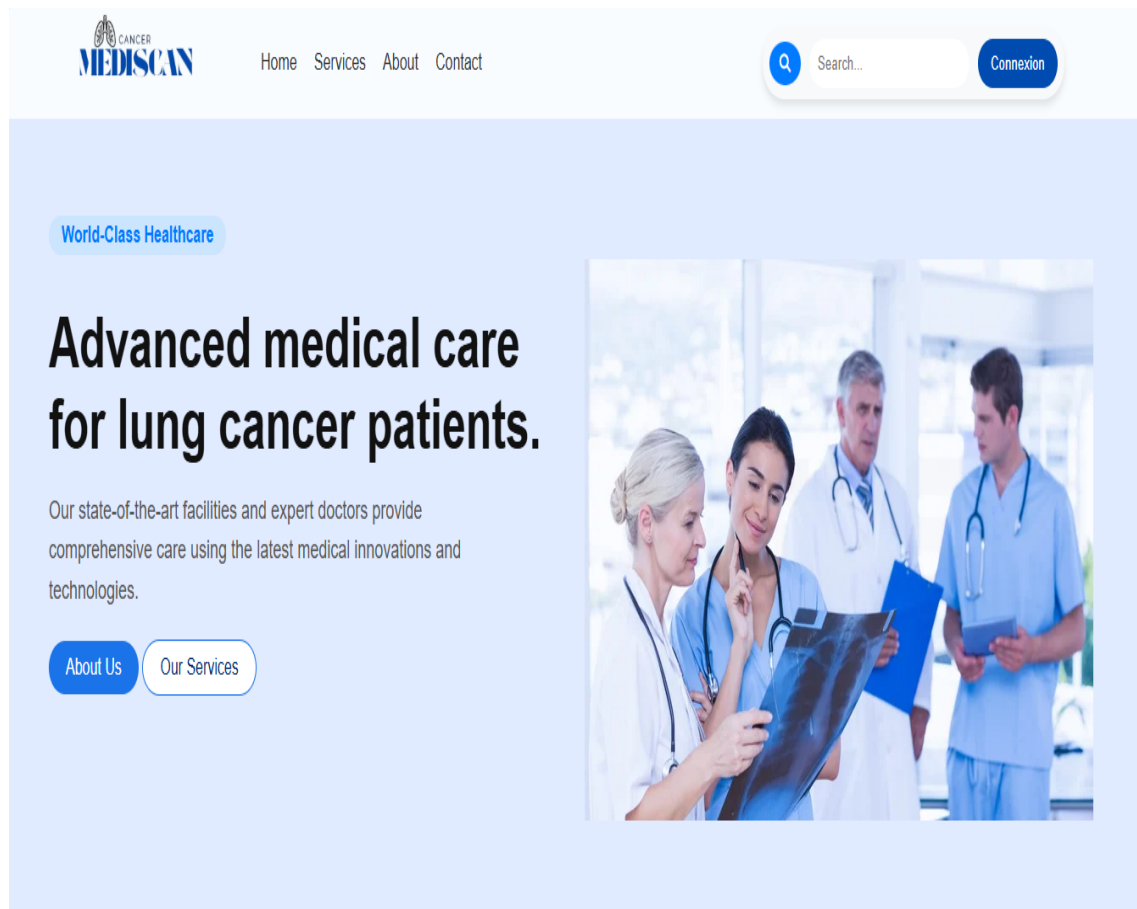


Figure 3.1: Home page

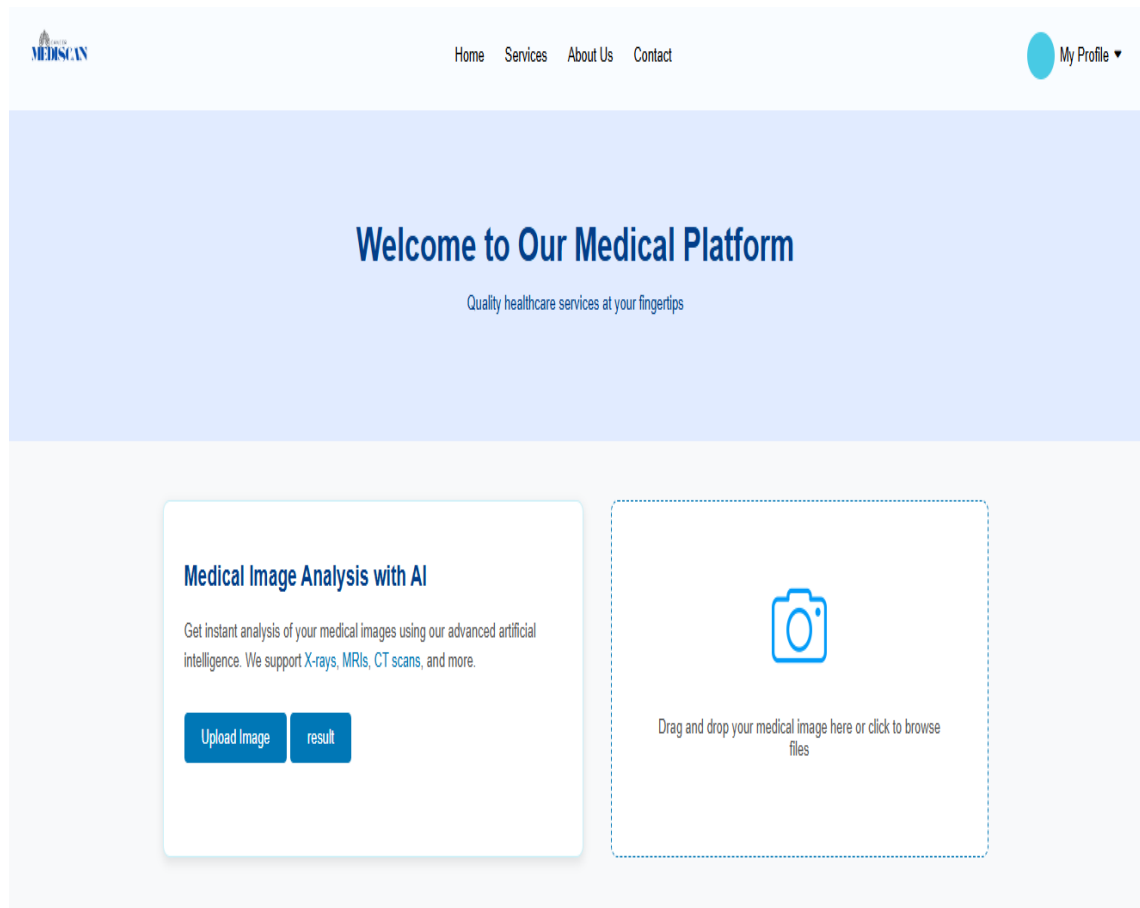


Figure 3.2: Scan Image page

Figure 3.3: book appointments page

This admin interface includes several important pages:

List Patients Page: Displays all registered patients in the system with the ability to edit or delete their accounts.

check Appointments Page: Allows reviewing all appointment requests submitted by patients and responding to them (confirming or rejecting).

list inquiries Page: Enables viewing and responding to messages and inquiries sent by patients through the website.

Manage Account Upgrades Page: Monitors patients who have made manual payments to upgrade their accounts, verifies the payment, and grants them the additional services once confirmed.

3.2 Part Two: Model Experimentation and Discussion

3.2.1 Experimentation

In this section, we will present that through the process of developing a multi-class medical image classification model, we conducted several consecutive experiments to achieve the

best model and improve its accuracy and efficiency. We will explain what was used in each experiment and the improvements made.

Experiment 1: In the first model, we trained the dataset directly without any preprocessing. We observed that the model's performance was very poor although the model was learning well during training, the validation accuracy was low, and the loss was high. This clearly indicated the presence of multiple issues, the most critical being overfitting.

Experiment 2: In this experiment, we applied preprocessing techniques to the dataset, such as resizing the images, removing noise, and addressing the class imbalance problem. We noticed an overall improvement in the model during training, with a gradual decrease in loss. This showed that the model was learning better than before and highlighted the importance of data preprocessing in enhancing the model's effectiveness.

Experiment 3: In this experiment, we performed advanced preprocessing of medical images using the Albumentations library. The applied transformations included: Resizing, Normalization, Horizontal Flip, Random Brightness and Contrast adjustments, and Rotation. These techniques are part of the data preprocessing phase and help improve the quality and diversity of the data before feeding it into the model.

Experiment 4: In this experiment, we used the TensorFlow library to create an efficient data pipeline using `tf.data.Dataset`. We divided the data into small batches during training (Batching), used prefetching to load the data in advance, and also applied shuffling to randomly mix the data. This helped accelerate the data loading process before feeding it into the training phase. As a result, we observed an increase in accuracy, a decrease in loss, and a faster training process. Also, the model design level we improved the architecture of the Convolutional Neural Network (CNN) by modifying the number of layers such as: Conv2D, MaxPooling2D, Flatten, Dense, and Dropout. We also changed the number of filters to (32, 64, 128) with a kernel size of 3×3 . In addition, we used the ReLU activation function in the hidden layers and Softmax in the output layer for multi-class classification. We observed a significant improvement in the model's performance compared to previous versions, as the validation accuracy increased and the loss value decreased significantly.

Experiment 5: In this experiment, we used the LIME technique to explain how the CNN model makes its decisions, since CNN is considered a black box model. LIME works by extracting an image from text and dividing it into small segments called superpixels. Then, it applies changes to these segments and uses the CNN again to make predictions, identifying the features that had the greatest impact on the model's decision. This technique helps us understand whether the CNN is making correct predictions or not. After applying LIME, we observed that the features it highlighted were located on the lungs,

such as opacity, masses, or other abnormalities, confirming that the model was focusing on the relevant areas when making its predictions.

Experiment 6: e built a deep learning model using Albumentations and TensorFlow libraries for multi-class classification for detection and we achieved good results . To validate our work and understand the impact of both libraries we created a secondary binary classification model, and the results were excellent.

3.2.1.1 Evaluation metrics and validation techniques

This table defines the difference between our fonal model and related experiments.

Model	Accuracy	Loss	Precision	Recall	F1-Score
Model 1	0.79	0.56	0.60	0.78	0.78
Model 2	0.80	0.47	0.82	0.81	0.80
Model 3	0.84	0.42	0.84	0.84	0.84
Model 4	0.9262	0.1942	0.93	0.92	0.93
Model 5	0.9262	0.9262	0.93	0.92	0.93
Model 6	0.9967	0.08	1.00	1.00	1.00

Table 3.3: Evaluation Results of Different Models

3.2.1.2 Comparison with other existing methods

This table defines the difference between our model and related work.

Model	Accuracy	Loss	F1-Score	Recall	Precision
[5]	21.81%	1.20	/	/	/
[4]	87.28%	/	0.72	77.56%	/
[10]	74.43%	/	/	74.96%	/
Model	0.88	0.45	0.87	0.86	0.88

Table 3.4: The difference between modele and some related work.

3.2.2 Discussion

Before developing our model for this work, we went through a series of steps across several successive experiments to improve the model's performance and accuracy. First, in the initial experiment, the model was trained on a raw and unprocessed dataset. The results were disastrous, with an accuracy of only 0.81 and a loss of 0.56. This poor performance was mainly due to the random data splitting, the imbalance in the number of images across classes, and the limited amount of data available. These issues highlighted the

necessity of applying proper data preprocessing techniques. In the second experiment, we applied a basic preprocessing techniques to address the issues identified in the first attempt. As a result, the performance reached a plateau, highlighting the need for more advanced and effective techniques to achieve further improvement.

In the third experiment, we improved the preprocessing stage by applying data augmentation using the Albumentations library. The results showed a noticeable improvement, with accuracy reaching 0.84, indicating the effectiveness of the technique.

In the fourth experiment, we aimed to speed up training to achieve better results by adding the tf.data technique from the TensorFlow library along with other improvements. For the model, the results were good, with accuracy reaching 0.92 and loss at 0.19. This demonstrates that this technique is highly effective.

In the fifth experiment, we aimed to understand how the model works and the reasons behind its class predictions. We added LIME as an explainer, and the results remained consistently good.

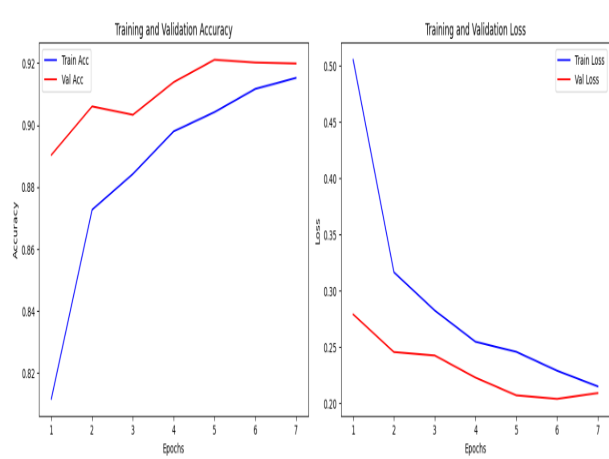


Figure 3.4: Training - validation curves

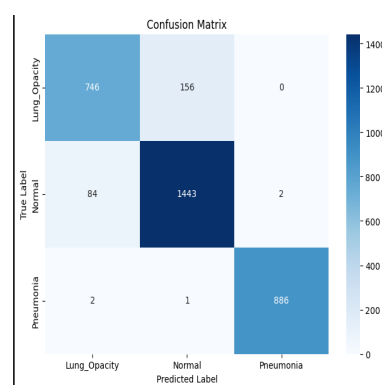


Figure 3.5: Confusion matrix

```

Classification Report:
              precision    recall  f1-score   support

Lung_Opacity   0.90      0.83      0.86       902
Normal         0.90      0.94      0.92      1529
Pneumonia      1.00      1.00      1.00       889

 accuracy              0.93      3320
 macro avg           0.93      0.92      0.93      3320
 weighted avg        0.93      0.93      0.93      3320
  
```

Figure 3.6: Confusion report

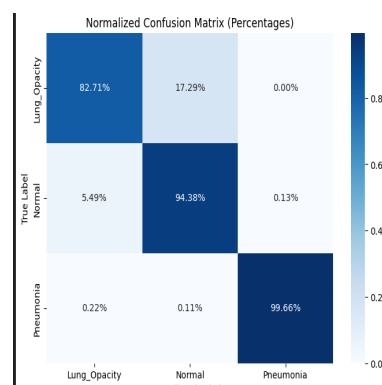


Figure 3.7: Confusion matrix percentage

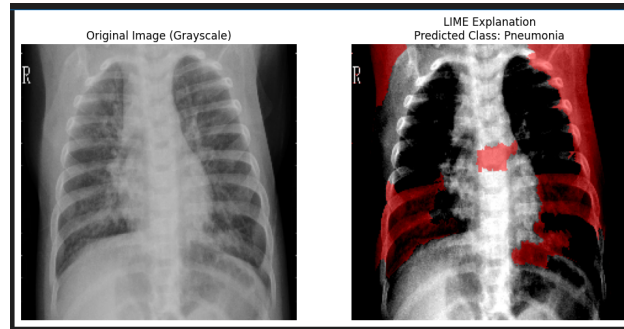


Figure 3.8: LIME explication

In the sixth experiment, to verify the model's effectiveness, the dataset was changed from multi-class to binary classification. The results were excellent, with accuracy reaching 0.99 and loss at 0.08. This demonstrates the model's effectiveness. However, the reason for the performance difference between the two tasks is attributed to the dataset quality, as there is some overlap between the "lung opacity" and "normal" classes, which leads to a relative decrease in accuracy in the multi-class setting.

Thus, we observe that the chosen model is effective for diagnosis and remains competitive compared to some previous works. However, there are some limitations affecting its performance, such as the quality and imbalance of the dataset.

Conclusion

This chapter presented the set the software and hardware used in the development of the work. It also provides an overview of the website's usage and how users interact with it. The second part details the various experiments we conducted to develop the appropriate model, as well as a comparison of our model with previous works. It concludes with a discussion and analysis of the results achieved.

General Conclusion



Synthesis

In this research, a deep learning strategy was proposed for the automatic diagnosis of lung cancer using chest X-ray images. The dataset includes three categories: Pneumonia, Lung Opacity, and Normal. The data was enhanced using the Albumentations library, and it was systematically split into training, validation, and testing sets using TensorFlow.

We relied on a CNN model for multi-class classification of diseases, achieving an accuracy of 92%, along with strong results in terms of precision, recall, F1-score, and support. Additionally, we integrated an interpretation tool for classification results to improve transparency and better understand the model's decisions. This model was then embedded in a web platform that includes pages for interaction between users and medical specialists.

Perspectives

Despite the positive results, the project has certain limitations, mainly the limited amount of data, which affects the model's generalization capability across all cases. Furthermore, the current website is limited to handling only lung cancer-related cases.

As part of future work, we plan to:

- Expand the data augmentation techniques.

- Add other diseases that can be diagnosed using artificial intelligence within the same platform, thus increasing the comprehensiveness and medical utility of the project.

Bibliography



- [1] American Cancer Society. What is lung cancer?, 2023. URL <https://www.cancer.org/cancer/lung-cancer/about/what-is.html>. Accessed: 2025-04-06.
- [2] Poonam J. Prasad and G.L. Bodhe. Trends in laboratory information management system. *Chemometrics and Intelligent Laboratory Systems*, 118:187–192, 2012. ISSN 0169-7439. doi: <https://doi.org/10.1016/j.chemolab.2012.07.001>. URL <https://www.sciencedirect.com/science/article/pii/S0169743912001438>.
- [3] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016. <http://www.deeplearningbook.org>.
- [4] Karam Karam, Houssein Chebbo, Sarah Saleh, Emanuel Youssef Dib, Elias Fiani, and Maroun Sadek. Medical reports.
- [5] Gianluca Accorsi, Giovanni Verri, Angela Acocella, David Saunders, Charlotte Martin de Fonjaudran, Diego Tamburini, Amarilli Rava, Samuel Whittaker, and Francesco Zerbetto. Retracted: Photophysics of artist’s pigments. *Measurement*, 123:293–297, 2018. ISSN 0263-2241. doi: <https://doi.org/10.1016/j.measurement.2018.03.082>. URL <https://www.sciencedirect.com/science/article/pii/S0263224118302756>.
- [6] RE Abdel-Aal. journal homepage: www.elsevier.com/locate/asoc. *Applied Soft Computing*, 11:5819–5828, 2011.
- [7] Clara Furió-Novejarque, Ricardo Sanz, Tobias KS Ritschel, Asbjørn Thode Reenberg, Ajenthen G Ranjan, Kirsten Nørgaard, José-Luis Díez, John Bagterp Jørgensen, and Jorge Bondia. Journal paper-computers in biology and medicine. *Medical Control Systems*, 154:107, 2023.

- [8] Kok Wei Khong, Ngozi Celestina Onyemeh, and Alain Yee-Loong Chong. Expert systems with applications. *Expert Systems with Applications*, 40:4858–4870, 2013.
- [9] Rabia Javed, Tahir Abbas, Ali Haider Khan, Ali Daud, Amal Bukhari, and Riad Alharbey. Deep learning for lungs cancer detection: a review. *Artificial Intelligence Review*, 57(8):197, 2024.
- [10] Worawate Ausawalaithong, Arjaree Thirach, Sanparith Marukatat, and Theerawit Wilaiprasitporn. Automatic lung cancer prediction from chest x-ray images using the deep learning approach. In *2018 11th Biomedical Engineering International Conference (BMEiCON)*, pages 1–5, 2018. doi: 10.1109/BMEiCON.2018.8609997.
- [11] Goram Mufarah M. Alshmrani, Qiang Ni, Richard Jiang, Haris Pervaiz, and Nada M. Elshennawy. A deep learning architecture for multi-class lung diseases classification using chest x-ray (cxr) images. *Alexandria Engineering Journal*, 64:923–935, 2023. ISSN 1110-0168. doi: <https://doi.org/10.1016/j.aej.2022.10.053>. URL <https://www.sciencedirect.com/science/article/pii/S1110016822007104>.
- [12] Jalal Rabbah, Mohammed Ridouani, and Larbi Hassouni. Improving pneumonia diagnosis with high-accuracy cnn-based chest x-ray image classification and integrated gradient. *Biomedical Signal Processing and Control*, 101:107239, 2025. ISSN 1746-8094. doi: <https://doi.org/10.1016/j.bspc.2024.107239>. URL <https://www.sciencedirect.com/science/article/pii/S1746809424012977>.
- [13] Hamzah Bumpenje, Rahmadwati, and Zainul Abidin. Chest cancer classification from chest ct-scan images using deep learning. In *2024 International Conference on Smart Computing, IoT and Machine Learning (SIML)*, pages 49–55, 2024. doi: 10.1109/SIML61815.2024.10578120.
- [14] Pathireddy Maheshwar Reddy MsB. N. Swarna Jyothi. Deep learning in oncology: Cnn based lung cancer detection using ct images. *International Journal of Innovative Research in Science Engineering and Technology*, 14(4), 2025.
- [15] Chamak Saha, Somak Saha, Md Asadur Rahman, M Haque Milu, Hiroki Higa, Mohd Abdur Rashid, and Nasim Ahmed. Lung-attnet: An attention mechanism based cnn architecture for lung cancer detection with federated learning. *IEEE Access*, 2025.
- [16] Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax dis-

eases. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3462–3471, 2017. doi: 10.1109/CVPR.2017.369.

- [17] Geert Litjens, Thijs Kooi, Babak Ehteshami Bejnordi, Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Mohsen Ghafoorian, Jeroen A.W.M. van der Laak, Bram van Ginneken, and Clara I. Sánchez. A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42:60–88, 2017. ISSN 1361-8415. doi: <https://doi.org/10.1016/j.media.2017.07.005>. URL <https://www.sciencedirect.com/science/article/pii/S1361841517301135>.